

Question	Answer	Mark	Mark scheme	Additional guidance
21	$x = -\frac{6}{7}, y = \frac{32}{7}$ $x = 2, y = -4$	M1 M1 M1	for substitution of a rearranged equation into a correct equation to form an equation in one variable. eg $3x^2 + 2(2 - 3x)^2 = 44$ or $3\left(\frac{2-y}{3}\right)^2 + 2y^2 = 44$ (dep on first M1) for multiplying out all brackets and collecting terms to form a simplified three term quadratic in any form of $ax^2 + bx + c (= 0)$ where at least 2 coefficients (a, b, c) are correct eg $21x^2 - 24x - 36 (= 0)$ or $7x^2 - 8x - 12 (= 0)$ or $7y^2 - 4y - 128 (= 0)$ (dep on first M1) for a suitable method to solve their 3 term quadratic using any correct method, for factorising, eg $(7x + 6)(x - 2)$ or $(7x + 6)(3x - 6)$ or $(21x + 18)(x - 2)$ or $(7y - 32)(y + 4)$ or correct use of formula, eg $\frac{8 \pm \sqrt{(-8)^2 - 4 \times 7 \times -12}}{2 \times 7}$ or $\frac{4 \pm \sqrt{(-4)^2 - 4 \times 7 \times -128}}{2 \times 7}$ or completing the square	Allow $(\pm 2 \pm 3x)$ for $(2 - 3x)$ (or $\left(\frac{\pm 2 \pm y}{3}\right)$ for $\left(\frac{2 - y}{3}\right)$) Implied by a correct equation (simplified or unsimplified) in terms of x or y eg $3x^2 + 2(4 - 12x + 9x^2) = 44$ or $3x^2 + 8 - 24x + 18x^2 = 44$ or $21x^2 - 24x = 36$ Look out for signs reversed The quadratic does not have to equal 0, ie accept $21x^2 - 24x = 36$ Can be implied by both x values or both y values correct (condone incorrect labelling) if the quadratic is correct If using the quadratic formula (condone one sign error, omission of brackets around the b in the $b^2 - 4ac$ and the fraction line not being under the $-$ in the $-b$. Allow some simplification – as far as eg $\frac{8 \pm \sqrt{64 + 336}}{14}$ or if factorising allow brackets which expand to give 2 out of 3 terms correct for their quadratic

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		M1	(dep on first M1) for substituting their 2 found values of x or y in a suitable equation or (dep on first M1) for one correct pair of values following from a correct quadratic	Condone substitution into their $(\pm 2 \pm 3x)$ or $\left(\frac{\pm 2 \pm y}{3}\right)$ Allow $-0.85(7\dots)$ or -0.86 for $-\frac{6}{7}$ Allow $4.57(1\dots)$ for $\frac{32}{7}$ If values of x or y are incorrect then working must be shown
		A1	for $x = -\frac{6}{7}$ oe, $y = \frac{32}{7}$ oe and $x = 2, y = -4$	Accept as coordinates Assume correct pairing unless clearly incorrect eg $\left(-\frac{6}{7}, -4\right), \left(2, \frac{32}{7}\right)$ Allow $-0.85(7\dots)$ or -0.86 for $-\frac{6}{7}$ Allow $4.57(1\dots)$ for $\frac{32}{7}$ If an answer is shown in the range in working and then incorrectly rounded award full marks A correct answer with no supportive working gets 0 marks